Ask Your Veterinanarian about Sildenafil to keep heart from beating like a big bass drum : The Happy Pill

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Sildenafil Seems Beneficial in Dog with Heartworm 8 years ago

Mike Marchywka*
157 Zachary Dr Talking Rock GA 30175 USA
(Dated: November 9, 2026)

This work describes the results giving sildenafil to one dog, Happy, who has had persistent cough since recovering from heartworm in 2019. She has been the topic of several recent manuscripts as her diet is adjusted to try to remove persistent cough. After an acute episode of apparent heart failure, she was treated with antibiotics and furosemdie before switching to sildenafil on request. Her recovery appears complete from the acute episode and her cough reduction is the most robust seen so far with even morning cough on initial wake up become reduced. Its likely her right sided heart and vasculature never healed properly from the initial heartworm infection. Some common medicines like pimodendan with positive inotropic effects may increase the problem as additional pressure results in more pathological hypertrophy or other adaptations. Sildenafil may improve pulmonary dilation and have a negative inotropic effect which may be therapeutic compared to pimobendan and ACE inhibitors for some dogs with heartworm related symptoms. Importantly however, she experienced an increase in coughing over 2 day which led to discontinuation of sildenafil due to concerns about worsening collapsed trachea. It may be worth consideration when heartworm damage is suspected without any echocardiography to confirm PAH but also to consider possibility of drug related side effects like trachea collapse.

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* marchywka@hotmail.com; to cite or credit this work, see bibtex in E

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I. INTRODUCTION

Heart failure is fairly common in dogs due to many reasons. Recently, new cases of DCM emerged in breeds that were not thought to be predisposed to the condition. Treatment for left sided failure such as degenerative mitral valve disease commonly consists of drugs such as furosemide, pimobendan, and an ACE inhbitor [36] . Pimobendan shows some early signs of beneficial left sided remodelling in myxomatous mitral valve disease (MMVD) [5] but here the question is considered of how this translates into heartworm related damage. Heartworm disease is another cause and is likely responsible for the chronic heart condition experienced by the dog discussed here, Happy. Her story has been detailed in several prior reports

but briefly she arrived here with heartworm in late 2018 and was treated with ivermection and doxycycline successfully. She briefly stopped coughing in early 2019 but continued to cough at varying amounts up until present in 2026. In response to acute episodes in the past, she was prescribed variations of the above with a diuretic, ACE inhbitors, and vetmeden.

In late 2020, another dog here, Andy, had partially recovered from severe heartworm disease after receiving Diroban on 2020-08-04 and 2020-09-08 but suffered from excessive abdominal fluid accumulation. He adopted out and, based on second hand information, a specialist diagnosed PAH based on echo and prescibed sildenafil which was thought to work well. Echo alone may not be complete [44] but given the heartworm history is probably quite convincing. On further investigation of heartworm specific heart damage it became clear that this may be a better approach for Happy than the prior drugs.

I.1. Heartworm

The damage caused by canine heartworm infection has been described as, " characterized by endothelial proliferation, perivascular inflammation and thrombo-embolism, all leading to chronic pulmonary hypertension and right-side congestive heart disease. " [9]. Little change in pulmonary parameters were noted during and 1 month after a 90 day heartworm treatment [39]. And indeed in the case of Happy little consistent change was observed in 7 years.

One well known target for this condition is PDE5 which appears to control nitric oxide levels in pulmonary vasodilation [11]. . Both PDE5 and PDE3 expression levels can increase in response to hypoxia [28]. Ventricular PDE5 contributes to remodelling in mice [45].

Some forms of pulmonary artery relaxation can be inhibited by heartworm infection although nitroglycerin response is intact even as some doubt about the mechanisms exist [27].

I.2. Right Ventricle Pulmonary Artery Death Spiral

Pylmonary hypertension can occur for a variety of reasons [3] although the immediate interest is PAH due to heartworm damage.

The two components contributing to PAH are several structural changes of muscle and matrix proteins as well as unregulated vasoconstriction [23].

Administration of sildenafil to hypoxic rats was able to reduce and reverse some pulmonary muscularization [38]/

The use of pimobendan and sildenafil in managing active adult heartworm infection has been discussed at least since 2019 where movement of worm from the heart was observed [43]. In that work, the positive inotropy of pimobendan was considered benefical in the acute situation of immediate concern. Simiarly, another work comparing pimobendan and sildenafil in chemically created PH found some benefits from both or in combination [30]. In a chronic setting, without resolution of PAH, it is essentially a right outflow obstruction. While beneficial in long term human heart failure studies, some excess deaths were noted in pimobendan treatment groups although not reaching statistical significance [10].

I.3. Sildenafil

It is available off-label in dogs and therefore specific prescribing information is not available. Literature reports suggest pharmacokinetics in dogs is highly variable in dogs with natural PH due to health state, other stomach contents, and hepatic metabolism which may saturate over the relevant oral dosage range of .5 to 3mg/kg (typically 2-3 times per day) [49] . Earlier work with induced PH from artificial empoli suggested increased half life at higher doses such as 4mg/kg versus 1mg/kg [3]. Dosing in the fed state decreased peak plasma concentration and AUC and again the non-proportional results were observed between 1 and 4mg/kg [2]. Accidental exposure of up to

14mg/kg have been survived according to informal reports [1]. Although daily dosing for 4months at 14.3mg/kg in PDE6A heterozygous dogs did produce reversible retina abnormalities and reductions in low light vision [18] . Although interestingly, rat pups recovered better from hypoxic retinal damage with sildenafil doses about 10mg/kg [17]. Concerns do exist about dangerout decreases in coronary artery flow when sildenafil is combined with nitrates [15].

In dogs with active heartworm infection that has migrated to the tricuspid valve, combined sildenafil and pimobendan in dogs without hypotension can cause worm to migrate to pulmonary artery [42].

In small breed dogs with various causes of PH the benefits of sildenafil were less clear when most dogs displayed syncope as well as cough [16].

One potentially serious side effect that may be relevant to dogs is the observation of desaturation in 2 of 269 pediatric patients leading to discontinuation of sildenafil and may be related to tracheobronchomalacia [13]. Effects related to obstructive sleep apnea are being explored although currently confusing [4].

The same targets, like TRPC members, may reduce smooth mucle proliferation but also cartilage sufficiency leading to some relief in hyperresponsivness but also adding to susceptibility to collapse. Sildenafil has a variety of effects on TRPC channel expression which may prevent pathological remodelling and contribute to modulation of airway responsiveness in rats [41]. Expression changes were also observed in pulmonary arterial smooth muscle [47] that may relate to its antiproliferative effect [46] . TRPC1 activity appears important to prevent chondrocyte senesence and proper cartilage growth [37]. It appears to improve bone fracture in mice in part due to angiogenesis [14] and delays pathological bone remodelling in aged mice [25]. Notably, it appeared to allow "superior" callus formation during healing in a rat fracture study [21].

Happy has been the subject of several recent works but limited progress had been made in curing her persistent cough which is believed to be due to a 2018 heartworm infection. Several dogs here have been treated for heartworm, including one pregnant pit bull. While all have survived, one dog, Andy, adopted out while still recovering in 2020. Based on second hand information, he was diagnosed with echo to have pulmonay arterial hypertension and prescribed Sildenafil. Upon further investigation, that appeared to be a good match for heartworm treated dogs with residual damage as it may reduce right side load selectively allowing for reverse remodelling and ultimate resolution. However, it was not until this acute episode that it could be evaluated on Happy over 5 years later.

When presented with heart failure, the prevailing standard of care appears to include furosemide, pimobendan, and an ACE inhibitor. Furosemide is a brute force attempt to remove excess fluid accumulation while the ACE inhibitor is largely to remove peripheral load. Pimobendan is largely a positive inotrope based on calcium sensitization but also had a PDE3 inhbiting effect possibly dilating pulmonary circulation. In the case of heartworm damage, which is predominantly right sided, its not clear how beneficial peripheral dilation would be. If there is heavy scarring in the pulmonary circulation dilation may not be significant and the positive inotropic effect on the right side may stimulate more maladaptive responses. Pimbdendan is strongly contraindicated in the presence of outflow obstructions but this warning may invoke largely concerns about the left side. Pimobendan

At least in rats with isoproterenol-induced heart damage, pimobendan had opposite deleterious effects on cardiac collagen while sildenafil appeared to benefit presumably due to difference in PDE3 and PDE5 effects [31].

Sildenafil is largely noted for PDE5 inhbitions and dilation of arteries controlled by this signalling. It also has a negative inotropic effect which may help allow reverse remodelling by eliminating the constant mechanical loading.

Thinking outloud



FIG. 1. Happy on 2020-12-16



FIG. 2. Happy on 2026-11-09. Due to differences in photo parameters, notably color balance or other processing, comparison to the 2020 image may be difficult but she is becoming more white on ventral side mostly.



FIG. 3. Happy's weight at various times. Claireś weight is shown for comparison generally measured on the same day. A household scale was used with significant tare weight possibly leading to fluctuations that were reduced later by allowing the scale to "warm up." Claire is of similar size as Happy but generally healthy although maybe a little heavier than she should be. They had diet adjustments and exercise that nominally tracked each other.

II. EVENT DESCRIPTION

Day	Serial	Date	Event
20604	0	2026-05-31	last day she ate ok
20607	3	2026-06-03	6 mg new furosmide started
20610	6	2026-06-06	began eating again
20611	7	2026-06-07	replace canola with soybean oil
20614	10	2026-06-10	first sildenafil 5 mg
20618	14	2026-06-14	final furosemide
20626	22	2026-06-22	slowly increase sildenafil to 6,10,10 mg
20641	37	2026-07-07	last snack until 18th, nomrinal continue sildenafil
20652	48	2026-07-18	restart snacks and regular early schedule
20652	48	2026-07-18	not coughing much
20657	53	2026-07-23	thought to be coughing more
20658	54	2026-07-24	clearly coughing a lot more but very active suspect trachea
20658	54	2026-07-24	last sildenafil
20735	141	2026-10-09	continues to forage well cough stable some neglect
20766	162	2026-11-09	cough, foraging good and stable little neglect

TABLE I.

Happy began to experience symptoms consistent with heart failure and was given expired furosemide, vetmeden, and an ACE inhbitors which had been presdribed for her years ago but not used. When she got to the vet, she was prescribed prednisone, furosemide, and "Amoxi-clav." Antibiotics and furosemide were given as prescribed but not the presnisone. She appareared to improve to almost her prior condition although remaining a bit slower than before and not quite as eager to eat. Upon request to Happy's vet, sildenafil was prescribed at 5mg 3x per day. She was found to do better with 6,10, and 10 mg and the furosemide was discontinued. Coughing seemed to improve although since she was a bit "slow" initially while recovering she is pretty much back to normal and coughing less on initial wake up and morning as well afternoon. Does not appear to cough as much during some stresses. She continues to be wary of some spoon feeding suspicious for dental pain or an unknown incident. As of 2026-07-08 or so she appears to continue to reduce coughing frequency and is generally better than she has been since 2019. Eventually her coughing increased greatly in 2day and the sildenafil was discontinued over concerns about collapsed traches mentioned above. She continues to "bounce around" indicating good energy and well being but the cough is loud and honky.

As of early October 2026, she continues to be playful and expressive with good foraging speed. She routinely plays with the bigger dogs Daisy and Murphy and barks while waiting for owner to return home. She appear to have some neglect however perhaps not responding all the time to food preperation cues. She is likely going deaf which may explain some of these issue. Recently she was sitting with her back towards me in the kitchen which previously was an indication something was wrong. She no longer hesistates to eat or prefers food dropped on floor to spoon (in the past suggesting something was wrong with her neck).

By early November 2026 seems to be stable in fairly low cough and medium speed mode with few obvious defects.

The switch to soybean oil was made due to concerns that the canola may have deleterious components. Dose fluctuated during the recovery period and a close look may be considered.

III. DISCUSSION

Pimodendan is strongly contraindicated in the pressnce of an outflow obstruction although some work with cats suggests it may be safe with some obstructions [48] . This is probably true for right sided obstructions as well as left. Scarred or hypertrophic pulmonary vasculature may continue to act as a right sided outflow obstruction even if some moderate dilation can be induced. Right sided positive inotropes would likely add stimuli for increases patholggical remodelling of both the heart and pulmonary system. While pimodendam apparently does have some PDE3 inhibiting ability, positive inotropic effects may be couterproductive. For the presumed heartworm related damage, without echo proven PAH, sildenafil may be a better guess to allow reverse remodelling and final resolution of heartworm maladaptations. Vascular remodelling is an incompletely understood process that seems to respond to details of the blood pressure waveform [8] and could be expected to respond to changes in ventricular waveform.

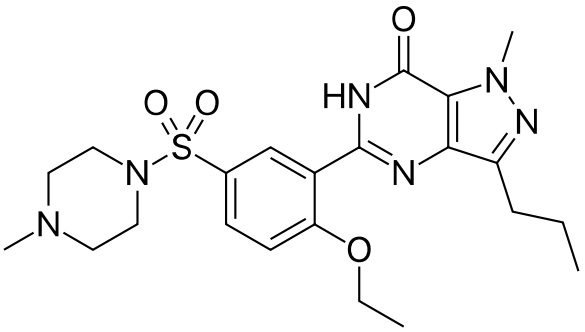


FIG. 4. Sildenafil, By Mrgreen71 - Own work, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=4994287 Believed to be accurate but not verified.

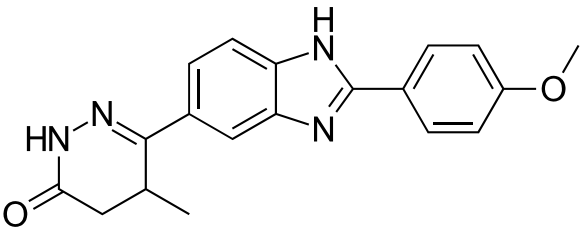


FIG. 5. Pimobendan, By Fvasconcellos - Own work, Public Domain, https://commons.wikimedia.org/w/index.php?curid=4511641 Believed to be accurate but not verified.

Happy is perhaps notable for a diet that is higher in copper and vitamin K than most dogs. In some cases, copper deficient diets however may reduce angioproliferation and PAH in response to some stimuli in rats but also stimulate right ventricle fibrosis [7] suggesting important tradeoffs and the importance of copper in heart health. It is possible in some diets that copper could be limiting in angiogenesis and healing from any worm related damage. It is thought that both of these supplementat contribute to her resilence although the initial event and clinical course without sildenafil and antibiotics remain unknown both of these could moderate disease severity through several mechanisms. In the past copper appeared to reduce coughing but as a sole focus has not entirely eliminated Happy's cough. Essential fatty acids may have been marginal too. Upon further investigation, vitamin D supplementation was not started until 2020-09-19 which was well after Happy's cough free week in early 2019. Andy had received several doses. Combined with high amino acid diet, there is some concern for excess muscle grwoth. Some role for vitamin D in muscle

growth generally is accepted [33] but details remain unclear and that is probably more true in dogs than humans. Interestingly, salmon supplemented with high dose vitamin D demonstrated increased cardiac experssion of a large ribosomal subunit [12] suggestive of increased ribosomal protein production. Indeed, overexpression of vitamin D receptor was found to produce skeletal muscle hypertrophy and ribosomal biogenesis [6]. It has been characterized as promoting VSMC growth and calcification [35].

Canola oil prior to the event was switched to soybean oil afterward and that may have contributed to some longer term effects. Essential and deleterious oil intake may be another factor to look at in more detail.

Previously, Happy seemed to be sensitive to tryptophan as well as other aromatic amino acids. Metabolites such as kynurenine may be elevated in PH [40] but also vasocilatory [29]. Tryptophan hydroxylase inhibitors are being investigrated to block serotonin synthesis with the potential for many beneficial effects on PAH and related remodelling [22] [26].

Other dietary components that may be relevant include argnine. From the above, nitric oxide production is a good suspect to mediate the clinical response and arginine may be a factor. Whlie peak amounts were as high as 250mg per sitting, which was higher than currently, average amounts were similar or lower. Its possible that slight elevations combined with other unidentified factors could act with or in place of sildenafil. One small study suggested that PAH patients metabolize arginine differently than healthy controls [19] which may be reducing NO synthesis and causing PAH although it could also be regulated by some other cause. In Happy's case the presumed problem is unhealed heartworm damage but mofidication of arginine metabolism throught diet may still be possible to encourage dilation. In the time between 2019 and 2026, she did have up to almost double the arginine amounts but with no remarkable changes.

Pimobendan appears to be a very good drug in specific applications but this has also made it trendy. While not intended for HCM it may still be beneficial in some respects. It appeared to have acute beneficial effects in HCM cats [32] although any reactions to chronic dosing would not be observed. It seems to be common for medicines, even blockbusters, to fix one immediate problem while possibly creating others at least in some populations. Pimobendan's most well known function is inotropy through calcium sensitization. It may be worth considering however if there is some reason for the heart to reduce calcium sensitivity before attmpting to enhance it. Copper deficiency is one common reason for hypertrophy and ultimate failure and there is some indication that deficient myocytes may be prone to death with normal calcium levels in ex vivo preparations although various channel blockers do not appear to improve the response [34]. Mice with heart specific loss of Ctr1 developed dilated cardiomyopathy and alterations in SERCA2 related regulators [20] further suggesting a link between Ca and Cu. Prior work discusses some issues with Cu and the heart [24] although not addressing Ca or pimobendan per se. Happy's diet was notable for copper supplementation averaging several mg per day because earlier she seemed to cough less and act better with more copper. Its possible that calcium desensitiation could be adaptive to some other condition that is possible fix. It would be interesting to see how many pimobendan responsive dogs do well with added copper in the right dietary context.

IV. CONCLUSIONS

For a dog with a history of heartworm and incompletely diagnosed heart problems, Sildenafil may be a better guess. Pimobendan may even be counter productive. The nature of heartworm damage is not completely elucidated but perhaps reduced signals for patholgoical remodelling can allow for reverse remodelling and final resolution. Its quite common in medicine to see one thing produce some improvements in similar conditions but lead to unappreciated damage in specific common situations. Pimobendan appears to be trendy but may not be a good match to the reality of many dogs' conditions. Echo and related tests can be expensive and time consuming and not even conclusive. "Talk to your vet about Viagra."

V. SUPPLEMENTAL INFORMATION

V.1. Computer Code

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2. Free software including Linux, R, LaTex etc.

3. Thanks everyone who contributed incidental support.

Appendix A: Statement of Conflicts

No specific funding was used in this effort and there are no relationships with others that could create a conflict of interest. I would like to develop these ideas further and have obvious bias towards making them appear successful. Barbara Cade, the dog owner, has worked in the pet food industry but this does not likely create a conflict. We have no interest in the makers of any of the products named in this work.

Appendix B: About the Authors and Facility

This work was performed at a dog rescue run by Barbara Cade and housed in rural Georgia. The author of this report ,Mike Marchywka, has a background in electrical engineering and has done extensive research using free online literature sources. I hope to find additional people interested in critically examining the results and verify that they can be reproduced effectively to treat other dogs.

Appendix C: Symbols, Abbreviations and Colloquialisms

TERM definition and meaning

Appendix D: General caveats and disclaimer

This document was created in the hope it will be interesting to someone including me by providing information about some topic that may include personal experience or a literature review or description of a speculative theory or idea. There is no assurance that the content of this work will be useful for any paricular purpose.

All statements in this document were true to the best of my knowledge at the time they were made and every attempt is made to assure they are not misleading or confusing. However, information provided by others and observations that can be manipulated by unknown causes ("gaslighting") may be misleading. Any use of this information should be preceded by validation including replication where feasible. Errors may enter into the final work at every step from conception and research to final editing.

Documents labelled "NOTES" or "not public" contain substantial informal or speculative content that may be terse and poorly edited or even sarcastic or profane. Documents labelled as "public" have generally been edited to be more coherent but probably have not been reviewed or proof read.

Generally non-public documents are labelled as such to avoid confusion and embarassment and should be read with that understanding.

Appendix E: Citing this as a tech report or white paper

Note: This is mostly manually entered and not assured to be error free.

This is tech report MJM-2026-002.

Version	Date	Comments
0.01	2026-07-09	Create from empty.tex template
-	November 9, 2026	version 0.00 MJM-2026-002
1.0	20xx-xx-xx	First revision for distribution

Released versions,

build script needs to include empty releases.tex

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Supporting files. Note that some dates,sizes, and md5's will change as this is rebuilt.

This really needs to include the data analysis code but right now it is auto generated picking up things from prior build in many cases

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